NLP Group Assignment 2

Combined technical report - Questions 1 to 4

DSL 504 - Natural Language Processing
Deadline: 30 October 2026, 11:59 PM IST
Coordinator and final submitter: Arnav Mishra

Member	ID	Primary question
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Section	Combined PDF pages	Content
Question 1	2-6	Models, curves, length analysis and six attention/error cases
Question 2	7-12	Six-page section; completed training/evaluation; seven comparative examples
Question 3	13-23	Revised experiment, generation attention, qualitative analysis and limitations
Question 4	24-29	Models, decoding, generated stories, quantitative results and limitations

The Q2 section is exactly six pages, including its figures and tables. There is no stated six-page cap for the whole group report. Question-specific section reports remain alongside the source; this PDF presents the complete four-question technical narrative. The question-level documents contain their experiment-specific references and limitations.

The contribution split and question responsibilities are listed in CONTRIBUTIONS.md.

Question 1: Attention-based English-Hindi translation

Owner: Rudra Mehul Dudhat. Two LSTM encoder-decoder systems compare fixed-state decoding with manually implemented additive attention. Results are documented by the executed notebook, saved histories, translations and attention figures.

1. Data preparation and architecture

The IIT Bombay corpus is normalized and tokenized, then empty/malformed pairs, excessive length/ratio and non-Devanagari targets are filtered. From 200,000 sampled raw training pairs, 50,000 cleaned pairs are used. Training pairs have at most 25 tokens per side; retained development/test pairs have at most 30. The filtered evaluation contains 430 development and 1,723 test pairs. Each language vocabulary contains 10,000 entries with PAD/SOS/EOS/UNK, fitted on training only. Maximum training length is 27 including boundary tokens; Hindi training OOV rate is about 9.8%.

Setting	Baseline	Attention
Architecture	128d embedding; 256d LSTM encoder/decoder; final state initializes decoder	Same sizes; additive scores over all encoder states; masked softmax context feeds decoder
Approx. parameters	5.9 million	6.3 million
Training (both)	Adam lr 0.002; batch 128; dropout 0.3; seed 0	8 epochs; teacher forcing; clip 1.0; validation-loss-selected checkpoint
Decoding (both)	Greedy; maximum 40 steps	PAD/SOS and UNK suppressed as disclosed

Attention uses v^T tanh(W_s s + W_h h_i), excludes padded source positions and forms a weighted encoder-state context. This exposes token-level weights without relying on a prebuilt translation model. The models share major training conditions, but attention adds parameters. Both were trained concurrently on CPU in the supplied run; epoch times are not isolated speed benchmarks.

Measured result	Baseline	Attention
Best validation loss	4.760	4.654
Teacher-forced validation next-token accuracy	0.289	0.302
Autoregressive test corpus BLEU (0-100)	0.60	0.92
Test word-overlap F1	0.187	0.205

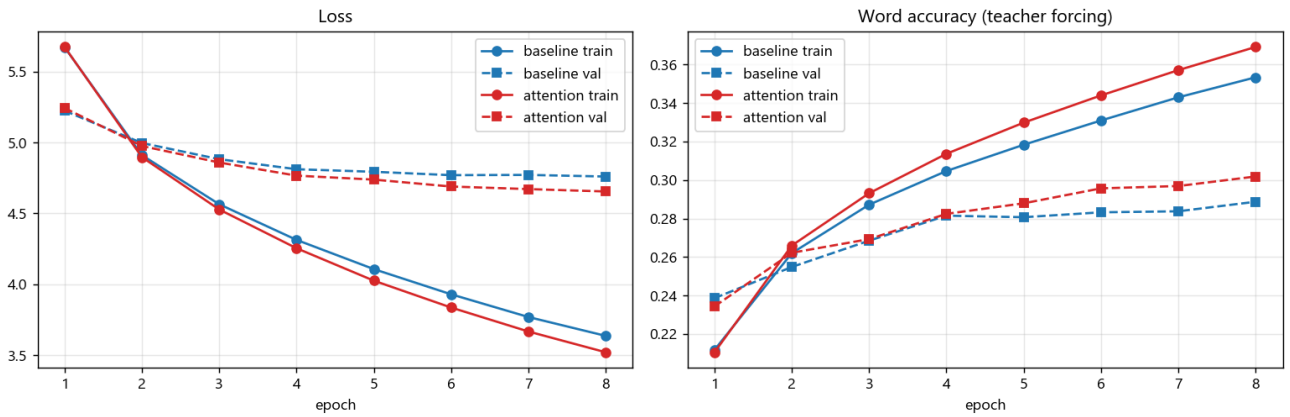


Figure Q1.1. Supplied training/validation losses and token accuracies. Teacher-forced accuracy supplies gold preceding tokens; it is not generated-translation accuracy. Both models remain weak under autoregressive decoding.

2. Length, errors and what attention explains

Source tokens	Test count	Baseline BLEU	Attention BLEU
1-5	10	1.97	2.50
6-10	256	1.25	1.14
11-15	455	0.56	1.09
16+	1,002	0.49	0.82

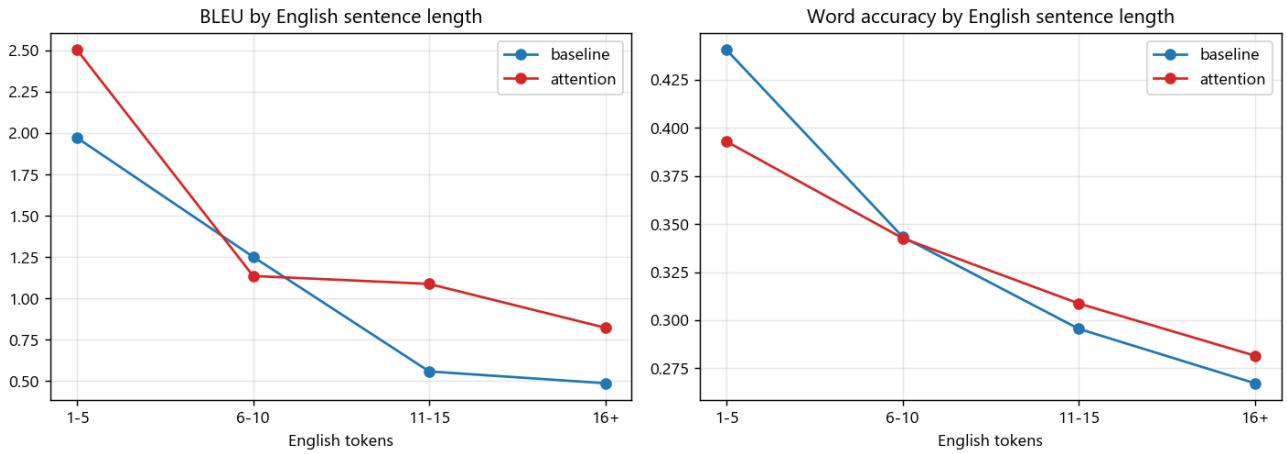


Figure Q1.2. Supplied length analysis. The accuracy subplot is teacher-forced next-token accuracy; BLEU uses generated translations. The shortest bucket has only ten sentences, so its apparent gain is uncertain. Attention improves BLEU in the 11-15 and 16+ buckets, while the 6-10 bucket is slightly worse. Word-overlap gains do not increase monotonically with length.

Attention improves word overlap on 926 test sentences, worsens it on 718 and ties on 79. Mean paired gain is 0.0178 with reported sentence-level standard error 0.0024; this is one seed per architecture and does not quantify training-seed uncertainty. Heuristic missing-word flags decrease from 43.0% to 38.2%, while repetition increases from 37.1% to 46.3% and extra-word flags from 29.5% to 34.3%. These rule-based categories are diagnostic indicators, not validated linguistic annotations.

The historical 30k-pair run used six epochs, vocabulary 8k and batch 64; the current run uses 50k/eight epochs/vocabulary 10k/batch 128. Its higher scores cannot be causally attributed to more data alone. The historical curves/translations are retained as supporting evidence rather than a controlled data-size ablation.

The supplied attention analysis reports a mean peak weight of 0.32, 86% of steps with no source position above 0.5 and sharpness/word-overlap correlation 0.20. Looking at a relevant source word can still yield an incorrect or incomplete translation. The first/last-third recall statistic concerns Hindi reference position; because word order differs across languages, it does not establish loss of the English source prefix. Likewise, a >=16-token input alone does not diagnose a long-distance dependency error. The six cases below describe observed omissions, substitutions and repetition without making that automatic causal claim.

Limitations: restricted noisy corpus, word-level vocabularies, small development set, only one model seed, greedy decoding and UNK suppression; the latter hides unknown-token outputs rather than recovering missing information. Neither BLEU below 1 nor visually sharp attention establishes useful translation. Longer training, subwords and other decoding are future options, not verified improvements in this run.

Sources: Q1 REPORT.md, executed notebook, saved histories, test_translations.csv and figures.

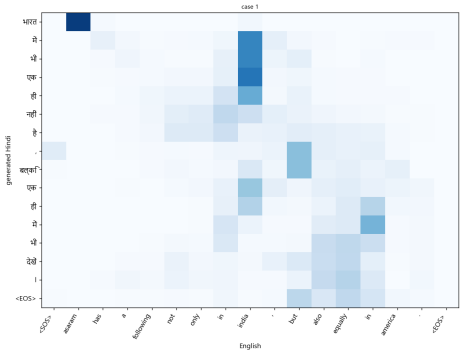
3. Attention error cases 1-2

Case 1. English: asaram has a following not only in india , but also equally in america .

Reference: आसाराम का भारत में ही नहीं , बल्कि अमेरिका में ही उतना भी प्रभुत्व है ।

Baseline: भारत में इस प्रकार की एक ऐसी हालत जो भी नहीं है ।

Attention: भारत में भी एक ही नहीं है , बल्कि एक ही में भी देखें ।



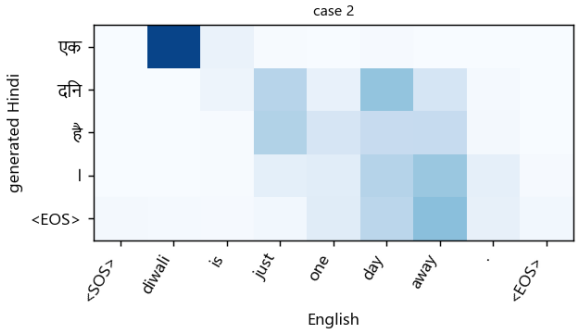
The unknown name Asaram is replaced; America is omitted. The contrastive connector is partly retained. Sharp focus on an OOV source name does not supply its target translation.

Case 2. English: diwali is just one day away .

Reference: दीपावली को एक दिन शेष बचा है ।

Baseline: एक व्यक्ति जो कुछ भी हो ।

Attention: एक दिन है ।



The generated text keeps one day but omits Diwali and the remaining-time relation. The model attends to an unknown event word while producing a generic phrase.

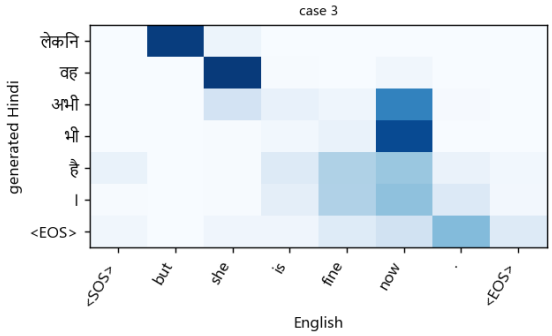
3. Attention error cases 3-4

Case 3. English: but she is fine now .

Reference: लेकिन अब वह अच्छी है ।

Baseline: लेकिन यह सच है ।

Attention: लेकिन वह अभी भी है ।



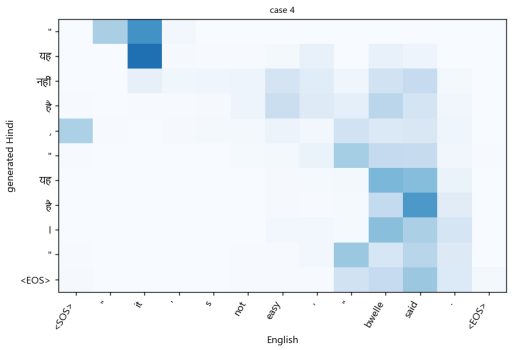
But/she/now are partly aligned, but the key property fine is absent. A plausible alignment is therefore insufficient for a complete translation.

Case 4. English: " it ' s not easy , " bwelle said .

Reference: वेल ने कहा , " यह आसान नहीं है । "

Baseline: " मैं तुम्हें नहीं है , " मैं तुम्हें नहीं है ।

Attention: " यह नहीं है , " यह है । "



The reporting name is unknown and output repeats fragments. Repetition and an incomplete quotation are directly observed; the heatmap is a clue rather than a causal proof.

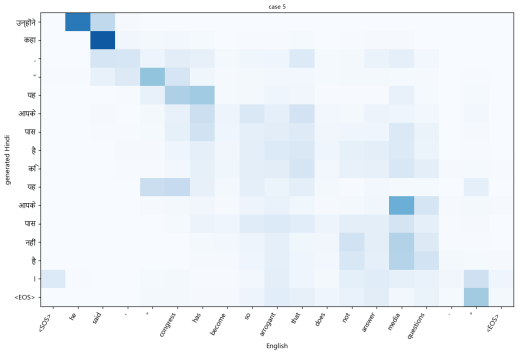
3. Attention error cases 5-6

Case 5. English: he said , " congress has become so arrogant that does not answer media questions . "

Reference: उन्होंने कहा , " कांग्रेस इतनी अहंकारी हो गई है कि वह मीडिया के सवालों का जवाब नहीं देती है । "

Baseline: उसने कहा , " ऐ मूसा ! " तुम कह रहा है , " तुम सच्चे हो , " तुम लोग गुमराही की

Attention: उन्होंने कहा , " यह आपके पास है कि यह आपके पास नहीं है ।



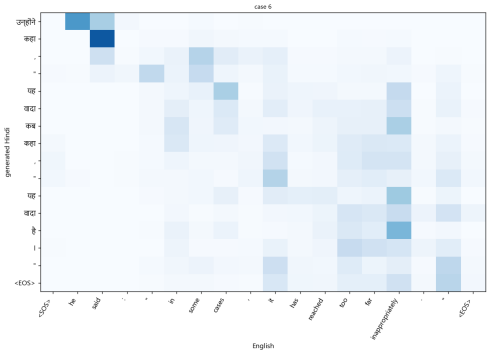
The reporting phrase survives, while the Congress/arrogance/media-answer content is lost or replaced. The long source alone is not proof of a specific dependency failure.

Case 6. English: he said : " in some cases , it has reached too far inappropriately . "

Reference: उन्होंने कहा किः " कुछ मामलों में , यह अनुपयुक्त रूप से बहुत आगे चली गई है । "

Baseline: उसने कहा , " हम तो एक ही एक अनुस्मृति है , " मेरे पास लाओ था ।

Attention: उन्होंने कहा , " यह वादा कब कहा , " यह वादा है । "



The reporting phrase survives but the quote becomes repetitive and semantically unrelated. OOV content and diffuse attention are plausible contributors, not isolated causes.

Question 2: Five-class Amazon review rating prediction

BiGRU with pretrained GloVe and additive attention versus fully fine-tuned RoBERTa with masked mean pooling. Experiments use seeds 42 and 123; first-token pooling is the selected ablation. This section reports inspected saved measurements, without additional model selection after final test evaluation.

1. Data, preprocessing and controlled training

The official SetFit English split contains 200,000 training, 5,000 validation and 5,000 test reviews, balanced across stars 1-5. SetFit supplies bodies but omits titles. Titles were recovered from the original-corpus mirror using exact agreement of ID, body, language and stars, then joined as title + newline + body for both models. Null titles (33 train, one validation) remain empty. Vocabulary and preprocessing fit training only; validation chooses checkpoints. Five exact train/validation text overlaps are disclosed while retaining the required official splits. Reserved test access followed a frozen protocol.

The classic pipeline HTML-unescapes, lowercases and removes tags/URLs, then applies regex tokenization. Minimum frequency 2 yields 32,611 vocabulary entries, capped at 60,000, with PAD/UNK and a 200-token limit. GloVe 6B 100-dimensional vectors cover 87.77% of vocabulary; uncovered vectors use fixed seed 2026. RoBERTa uses original text, byte-level BPE and a 256-token limit with dynamic padding to multiples of eight. Dataset/model revisions and hashes are pinned in config.json and manifests.

Setting	BiGRU + attention	RoBERTa mean / first
Architecture	2 BiGRU layers; 128 hidden/direction; additive attention width 128	roberta-base; all encoder weights trainable; masked mean / first token
Head / parameters	Dropout 0.3; linear -> 5; 3,768,497	Dropout 0.2; linear -> 5; 124,058,885
Optimizer	Adam, learning rate 0.001	AdamW, learning rate 0.00002; decay 0.01; 8% warmup then linear decay
Batch / precision	128; float32	Global microbatch 16; accumulation 2; effective 32; recorded bfloat16
Selection / cap	Validation Macro F1; patience 2; at most 8 epochs	Validation Macro F1; patience 1; at most 3 epochs

Both use cross-entropy, gradient clipping 1.0 and minimum improvement 0.0001. GloVe is frozen for epoch one then unfrozen. Packed GRU sequences and masked attention exclude padding. RoBERTa mean pooling includes all non-padding states, including special tokens. AdamW excludes biases/LayerNorm from decay. All recorded runs used Tesla T4 hardware; optimizer/scheduler state was retained during active recovery.

Run	Seed	Epochs / best	Training h	Execution
baseline	42	7 / 5	0.22	1 GPU(s)
baseline	123	6 / 4	0.19	1 GPU(s)
roberta_mean	42	3 / 3	6.44	1 GPU(s)
roberta_mean	123	3 / 2	3.55	2 GPU(s)
roberta_first	42	3 / 3	3.56	2 GPU(s)
roberta_first	123	3 / 2	3.43	2 GPU(s)

2. Learning curves and selection

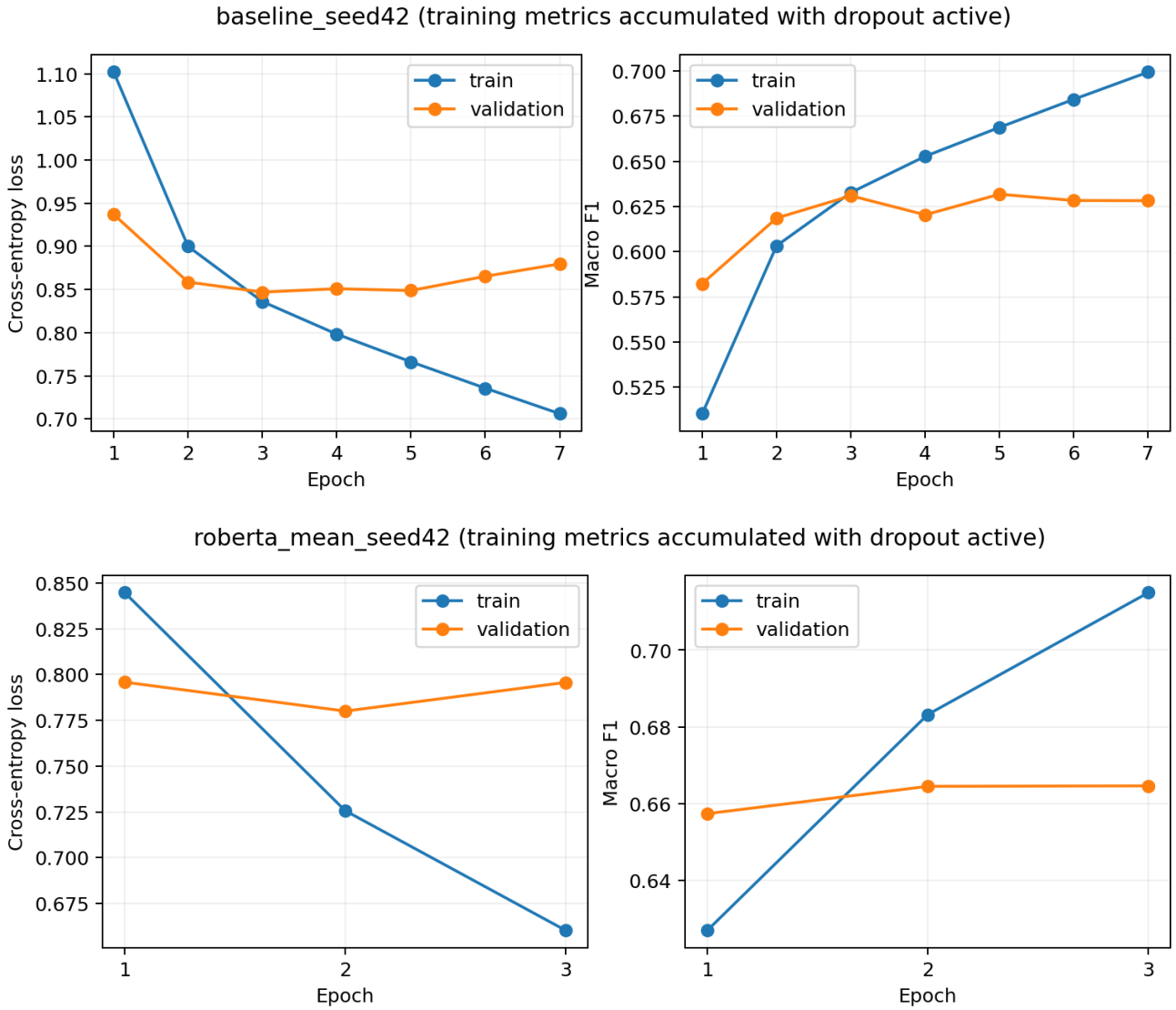


Figure 1. Training/validation loss and Macro F1 for seed 42 of the two main models. Equivalent figures for seed 123 and both ablation seeds are supplied in results/figures/. Training metrics accumulate with dropout active; validation uses evaluation mode, so the two curves are not identical evaluation protocols.

Baseline best epochs were 5/4 for seeds 42/123, with early stopping at epochs 7/6. Mean-pooling RoBERTa selected epochs 3/2; both runs completed the three-epoch cap. Checkpoints use validation Macro F1, not minimum loss or test accuracy. Full histories retain the actual losses and scores rather than only the selected epochs.

3. Frozen final test evaluation

Each main model/seed produced one final 5,000-review test prediction file. Inspection checked sample IDs, probability ranges/sums, argmax labels and recomputed metrics, including aggregate means and sample SD (ddof=1). Predictions use argmax with labels 0-4 mapped to stars 1-5. First-token models are evaluated on validation for the ablation, not additionally on test.

Model	Accuracy (%)	Macro F1	MAE
BiGRU	63.79 +/- 0.78	0.6351 +/- 0.0069	0.4277 +/- 0.0075
RoBERTa mean	67.13 +/- 0.04	0.6706 +/- 0.0010	0.3672 +/- 0.0000

RoBERTa improves accuracy by 3.34 percentage points and Macro F1 by 0.0355, while reducing MAE by 0.0605 stars. These are descriptive two-seed comparisons, not confidence intervals or significance tests. Smaller observed seed variability does not establish robustness.

True rating	BiGRU F1, mean +/- SD	RoBERTa F1, mean +/- SD
1	0.7197 +/- 0.0032	0.7454 +/- 0.0085
2	0.5483 +/- 0.0072	0.5911 +/- 0.0080
3	0.5360 +/- 0.0107	0.5792 +/- 0.0059
4	0.5908 +/- 0.0129	0.6369 +/- 0.0002
5	0.7805 +/- 0.0007	0.8004 +/- 0.0055

RoBERTa improves F1 for every rating; the largest gain is for four stars (about 0.0460). Both models identify extreme ratings more accurately than two-, three- and four-star reviews. Middle-rating wording is often mixed and neighboring rating labels can be ambiguous.

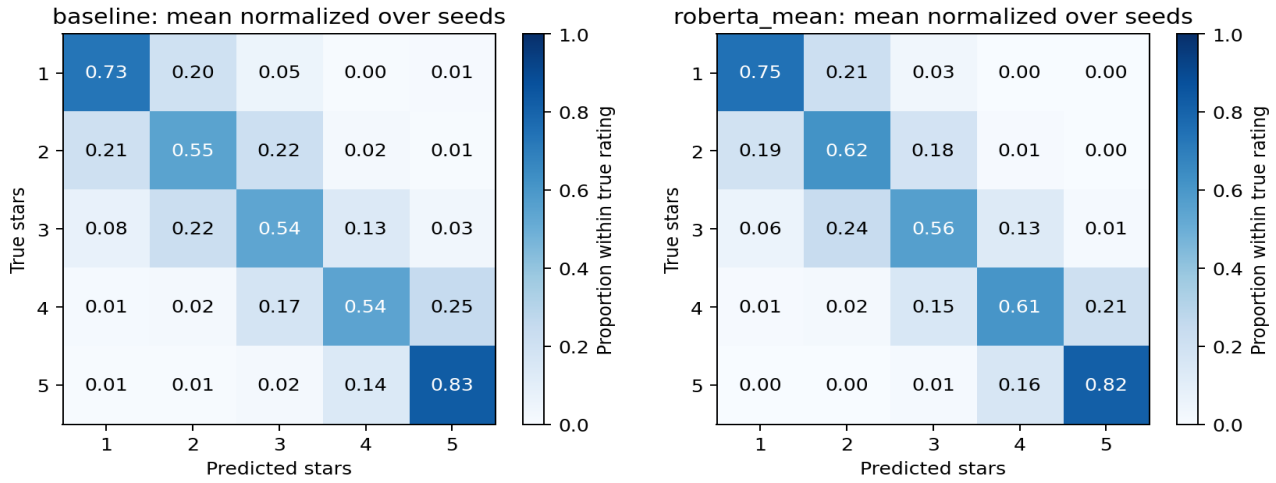
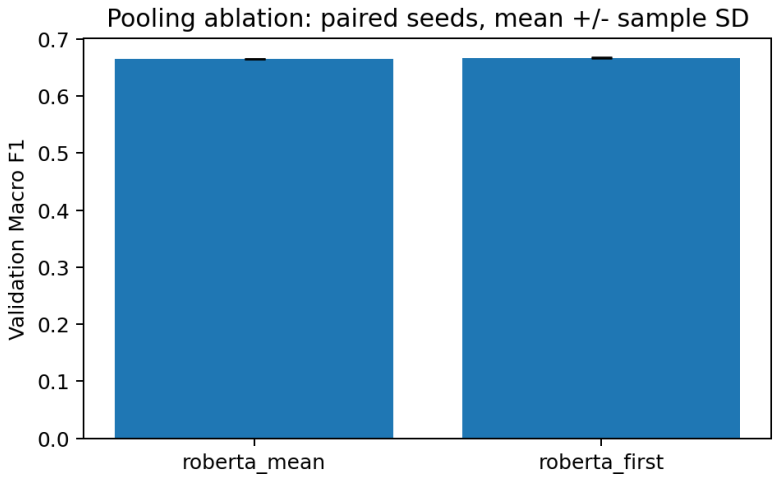


Figure 2. Row-normalized test confusion matrices averaged over seeds: BiGRU (left), RoBERTa (right). Full seed-specific matrices are supplied.

Adjacent-rating error rates fall from 30.84% to 29.49% of all reviews; distant errors (distance >=2) fall from 5.37% to 3.38%. Adjacent errors comprise 85.16% versus 89.72% of mistakes: the larger share for RoBERTa reflects fewer distant errors, not a higher overall adjacent-error rate. Extreme 1-versus-5 error rates fall from 0.29% to 0.09%.

4. Pooling ablation, reproducibility and limitations

Validation result	Mean pooling	First-token pooling
Macro F1 mean +/- SD	0.664875 +/- 0.000313	0.666848 +/- 0.000459
MAE mean +/- SD	0.3738 +/- 0.0006	0.3725 +/- 0.0016



First-token pooling is slightly better on validation in both seeds: mean-minus-first Macro F1 differences are -0.001869 (seed 42) and -0.002076 (seed 123). Its average MAE is lower by 0.0013. These small differences do not establish a universal pooling advantage. The required mean-pooling architecture remains the main final-test model; the ablation does not trigger test-driven architecture replacement.

Execution topology limits pooling-only attribution. Mean seed 42 used one GPU, while mean seed 123 and both first-token runs used two-GPU DistributedDataParallel. Global microbatch 16, accumulation 2 and effective batch 32 were preserved; gradient/communication probes passed, but rank-specific dropout streams and numerical reduction order can alter trajectories. The seed-123 pair shares topology; the seed-42 pair does not. Both baselines used one GPU. Actual topology, precision and source provenance are retained per run.

Training and evaluation were performed in the recorded Kaggle environment; CPU finalization only inspected and packaged saved files. It did not retrain or access the test corpus. Six training archives retain the selected checkpoints; the compact submission component excludes full data/caches/checkpoints. The final test protocol hash is recorded in FROZEN_PROTOCOL.json, and generated analysis can reuse its saved predictions.

Limitations include only two seeds, a single review domain, rare/OOV tokens, bounded sequence lengths, five known train/validation text overlaps, and ambiguity between neighboring star labels. RoBERTa also benefits from extensive external pretraining, so this comparison does not isolate architecture alone. No calibration, sarcasm understanding, deployment safety or out-of-domain robustness was established. Improvements should be selected on new training/validation experiments, not the final test results.

Conclusion: RoBERTa mean pooling improves the measured rating-classification and ordinal-error metrics over this GloVe-BiGRU baseline. First-token pooling has a small validation advantage in the present ablation. Neither finding establishes a general superiority across datasets, execution environments or untested hyperparameters.

5. Baseline-wrong / RoBERTa-right examples

Fixed seed 42, selected by ascending review ID: five of 500 available baseline-wrong/RoBERTa-right cases. Labels and predictions are unedited; interpretation below is a text-supported hypothesis, not a causal account of internal reasoning. Full texts are also retained in qualitative_examples.csv.

en_0002917: true 3; BiGRU 4; RoBERTa 3

Title: I also think I need to try a different coffee because I haven't been real pleased with the flavor of the coffee
Review: I've done a few so far but getting the hang of securing the foil to the capsule takes patience and practice. I've put holes in a few foils and had to redo them. I also think I need to try a different coffee because I haven't been real pleased with the flavor of the coffee. Also, the original foils on the capsules can be difficult to remove.

Mixed sentiment and practical difficulties. The review combines partial progress with flavor dissatisfaction, damaged foils and difficult capsule preparation. The neutral three-star label fits this mixed evidence. RoBERTa may better combine these qualifications; the positive four-star baseline prediction underweights complaints.

en_0003912: true 3; BiGRU 2; RoBERTa 3

Title: I won’t wear it much
Review: It’s a bit expensive and doesn’t lay right, it sticks to everything.

Rating ambiguity and malformed text. The text is strongly critical, but its recorded label is three stars. RoBERTa matches that label while the baseline predicts two. This is label agreement, not proof that three stars is the uniquely correct interpretation. The malformed apostrophes are present in the saved text.

en_0005862: true 2; BiGRU 3; RoBERTa 2

Title: Loved this purse but
Review: Loved this purse but, I can't take the bad odor coming from inside the purse. I even aired it out for two days and there was no difference. In process of returning.

Contrast and overriding complaint. The positive opening is reversed by persistent odor, unsuccessful airing and a return. RoBERTa predicts two stars while the baseline predicts three; sensitivity to the contrast and return intent is a plausible explanation, not a demonstrated reasoning process.

en_0008881: true 2; BiGRU 3; RoBERTa 2

Title: ... Expected it to be made out of a material like a bath towel
Review: Too Warm - Expected it to be made out of a material like a bath towel. Very soft, but seems like an old lady would wear this to stay warm inside a building that the A/C is on in the summertime on a 95 degree day...

Mixed attributes and unmet expectations. Softness is positive, but excessive warmth and the mismatch with expected towel-like material dominate the complaint. RoBERTa predicts two stars; the baseline predicts three. This suggests different weighting of positive attributes and dissatisfaction.

en_0009772: true 2; BiGRU 3; RoBERTa 2

Title: Could not find the answers I needed
Review: The Puzzle dictionary is too heavy to handle. The words, Clues, I tried to look up were more difficult than the clue itself. Would not suggest it for making Crossword Puzzles easier.

Negative utility and recommendation. The dictionary is described as heavy, unhelpful and difficult to use, with an explicit negative recommendation. RoBERTa matches two stars while the baseline gives three. This example concerns failure of the intended use rather than sarcasm.

6. Reverse cases and token-level importance

Two of 362 available baseline-right/RoBERTa-wrong cases, again in ascending review-ID order for seed 42. The stronger aggregate model still fails on individual reviews.

en_0000916: true 3; BiGRU 3; RoBERTa 2

Title: Nice mat but slips all over the place
Review: Doesn't stay put. Annoying.

Positive title versus short negative body. The title calls the mat nice but the body says it slips and is annoying. The baseline matches three stars; RoBERTa predicts two. The latter may overemphasize the complaint. The short mixed review illustrates uncertainty in mapping wording to exact stars.

en_0005599: true 5; BiGRU 5; RoBERTa 4

Title: Very classy and very sturdy
Review: We bought this to use in our RV. We like it very much. Very classy and very sturdy (so far). It is not light weight (ie, like Correlle) but we were not necessarily looking for anything lightweight). However, is is very attractive and so far we have enjoyed using it in our RV.

Qualification within strong approval. The reviewer repeatedly likes the product and accepts its weight as compatible with their needs. The baseline matches five stars; RoBERTa gives four. Mention of weight or the qualification so far may contribute, but the explanation remains a hypothesis.

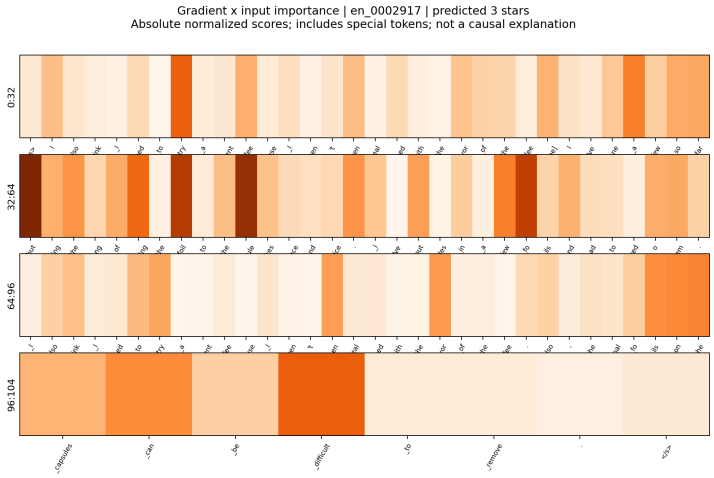


Figure 3. Absolute gradient x input importance for the predicted-class logit of the fixed RoBERTa model on en_0002917. Scores are normalized across all retained tokens, including special tokens. The displayed subset is not the complete review; full scores are in token_importance.csv.

In the saved scores, the contrast marker but (position 32) has importance 0.0398; capsule 0.0368, foil 0.0334, the fo subword of foils 0.0322, difficult 0.0264 and try 0.0261 also receive substantial weight. This sensitivity is consistent with practical difficulties and mixed sentiment, but does not establish how the model combined them or prove negation understanding. Tokens for the first occurrence of the negation have smaller scores than some context tokens. The example has 104 retained tokens, below the 256-token limit; truncation is therefore not the explanation for this particular difference. Other long reviews may lose text.

Reproduction/evidence: question_2/README.md and KAGGLE_RUN_GUIDE.md describe exact phases; config.json, manifests, environment records, training_records/, final prediction CSVs and metric JSONs retain run-level provenance. Qualitative interpretations are based on the saved review text and model predictions.

Sources: Assignment PDF pp. 5-7; SetFit/amazon_reviews_multi_en and buruzaemon original-corpus mirror (exact IDs/revisions in config.json); Stanford GloVe 6B; FacebookAI/roberta-base. Full source links and recorded settings are retained in the component README/configuration.

Abstractive Text Summarization

Question 3 - From-scratch Transformer Encoder-Decoder

Measured CPU experiment on CNN / Daily Mail 3.0.0. All reported model scores come from autoregressive generation with the validation-selected checkpoint; no pre-trained model weights are used.

1. Data and preprocessing

The official splits contain 287,113 training, 13,368 validation, and 11,490 test pairs. To fit CPU resources, this experiment uses the first 2,048 training pairs and first 1,024 validation pairs in repository order. The training subset is 0.71% of the full training split and is not a random representative sample. All 11,490 test articles are evaluated. No test example trains the tokenizer, model, or hyperparameters.

SentencePiece BPE is trained exclusively on lowercased articles and highlights from the training subset. One shared 4,096-token vocabulary is used for both sides, including PAD=0, UNK=1, BOS=2 and EOS=3. SentencePiece applies NFKC normalization. Source articles are truncated to 400 subwords. Summaries have at most 128 tokens including BOS and EOS (126 content subwords). Batches are padded to their longest sequence. Original reference highlight newlines are retained separately for scoring.

2. Architecture

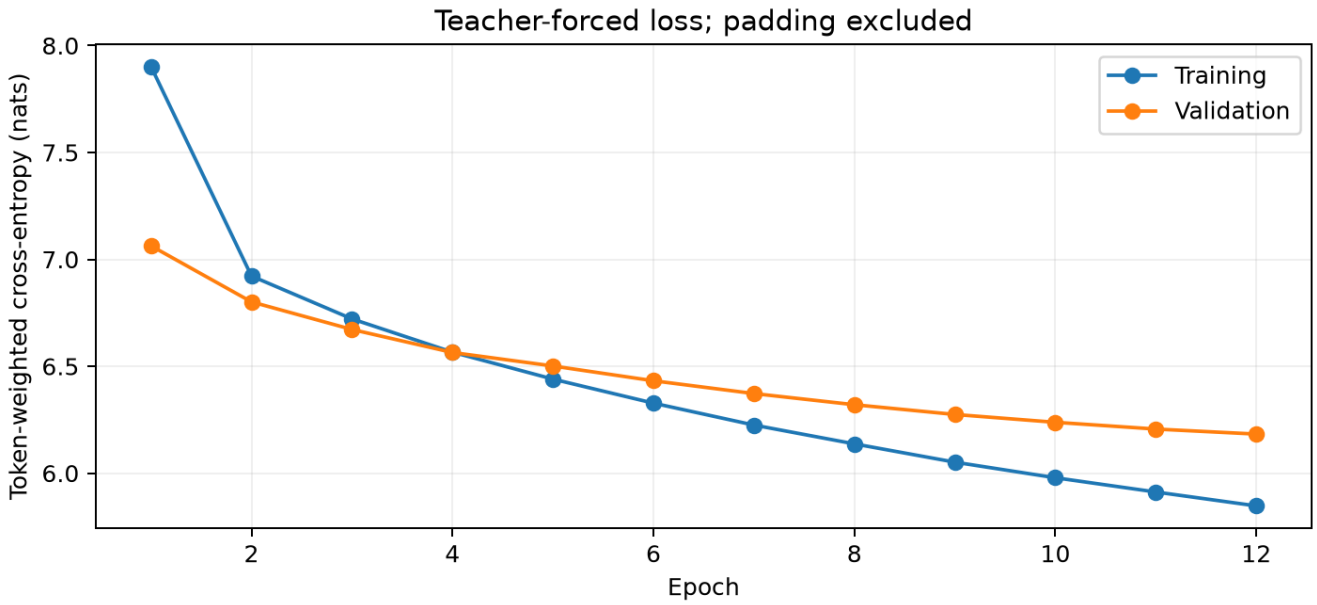
Setting	Value
Encoder / decoder layers	2 / 2
Model / feed-forward width	64 / 256
Attention heads	4 (16 dimensions each)
Dropout / normalization	0.1 / post-residual LayerNorm
Trainable parameters	761,856
Position / output layer	Sinusoidal encoding / linear vocabulary logits

Query, key and value projections are reshaped into heads. Attention is softmax(QK^T / sqrt(16) + mask)V, followed by concatenation and a linear projection. Feed-forward layers use Linear -> ReLU -> Dropout -> Linear. Each sublayer uses LayerNorm(x + Dropout(sublayer(x))). The source padding mask is used in the encoder and cross-attention. Decoder self-attention combines a triangular causal mask and target-key padding mask. All components use elementary tensor operations and basic neural layers.

3. Training and checkpoint selection

Teacher forcing uses gold summaries shifted right as decoder inputs and shifted left as targets. Cross-entropy ignores PAD; epoch losses are weighted by the number of non-padding target tokens. Adam uses betas=(0.9, 0.98), epsilon=1e-9, gradient clipping at 1.0, batch size 16 and seed 42. The learning rate is 0.1 * 64^(-0.5) * min(step^(-0.5), step * 200^(-1.5)). Early stopping has patience 2 after epoch-end validation; the best checkpoint is restored.

The run completed 12 epochs. Best epoch: 12; validation loss: 6.1827 nats/token. Training time: 15.7 minutes on CPU with 4 PyTorch threads. PyTorch 2.14.1+cpu. This width and depth are intentionally smaller than the suggested example configuration because no GPU is available.



Selected epochs are tabulated below; all epochs and learning rates are available in results/history.csv.

Epoch	Training loss	Validation loss
1	7.9018	7.0632
3	6.7215	6.6720
5	6.4398	6.5012
7	6.2250	6.3722
9	6.0514	6.2745
11	5.9118	6.2062
12	5.8469	6.1827

4. Full-test quantitative evaluation

Greedy decoding begins with BOS, feeds predicted tokens back into the decoder, and stops at EOS or 128 generated tokens. PAD and BOS are blocked as output tokens. There is no reference input at generation time. Each article is encoded once; projected decoder keys and values are cached. Caching changes computation cost, not the greedy output (verified by tests).

Scores below are macro averages of per-example F1 multiplied by 100 on all 11,490 test pairs using rouge_score with use_stemmer=True. ROUGE-L uses one sequence; ROUGE-Lsum uses sentence-newline boundaries. Generated text and Lead-3 use an offline punctuation-based sentence heuristic; reference highlights preserve their existing newlines. This heuristic can split abbreviations incorrectly and is a limitation.

System	R-1	R-2	R-L	R-Lsum
Transformer	1.75	0.05	1.63	1.70
Lead-3 (measured)	39.91	17.38	24.90	36.20
Lead-3 (assignment)	40.06	17.48	25.02	36.36

Lead-3 copies the first three sentences of each full original article. It has access to the full article, whereas the model sees only its first 400 subwords. Its measured scores may differ from the assignment reference because sentence segmentation is heuristic. Both systems are scored with the identical local metric configuration.

Transformer minus measured Lead-3 ROUGE-1: -38.16 points. Learning fluent, factual generation from only 2,048 examples is much harder than copying informative news leads. The 4,096-piece vocabulary also splits rare names and numbers into many fragments, and source truncation can remove important details.

There are 4,190 distinct decoded predictions across the 11,490 inputs. Distinct outputs do not imply good summaries: repetition, omitted facts and unsupported statements must still be assessed. The original run collapsed to one output. The revised run reduces the learning-rate scale from 1.0 to 0.1 following validation-only diagnostics. The model architecture, training subset, tokenizer and seed are unchanged. The checkpoint is selected by validation loss, before final test scoring. The original test set has already been evaluated, so this is a reused benchmark, not a new blind held-out experiment. No test metric selects the revised hyperparameters or checkpoint.

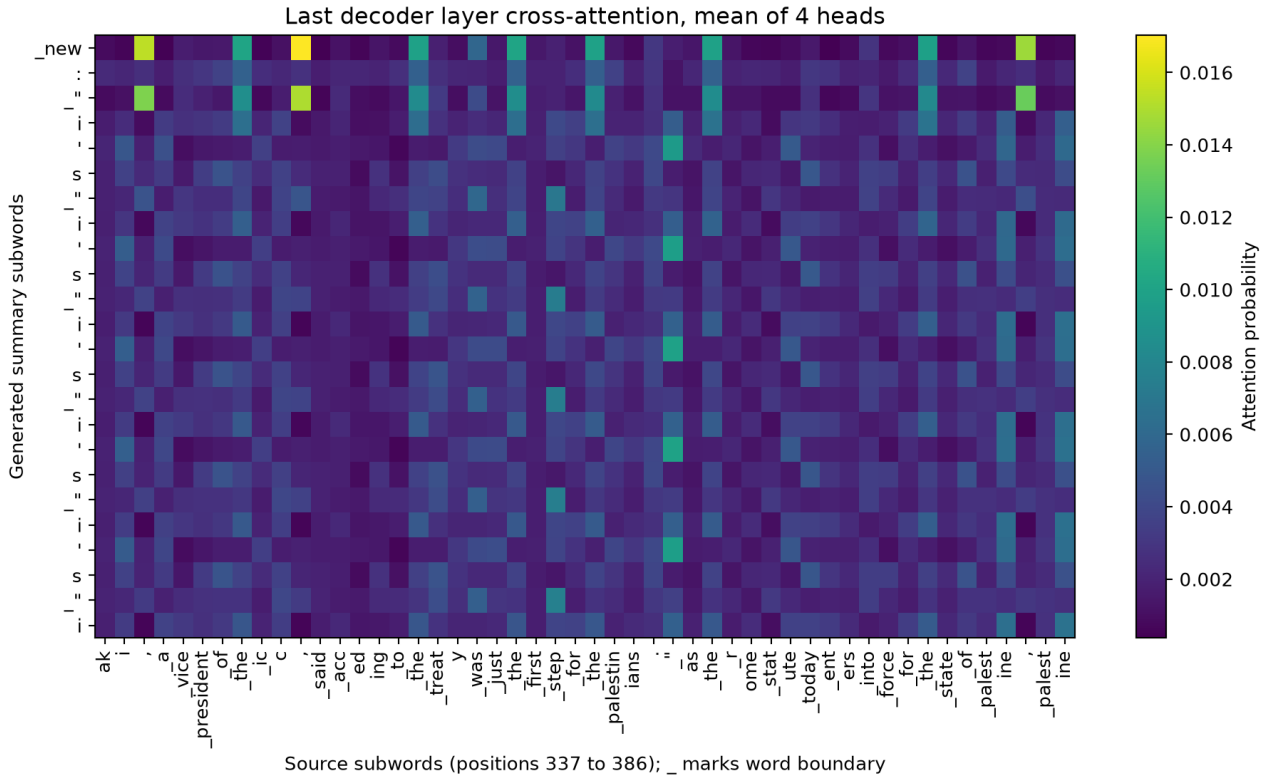
EOS-terminated outputs: 3,964/11,490. Outputs with any repeated whitespace trigram: 11,036/11,490. Generation plus scoring took 2.8 minutes. Repetition is measured on unedited decoded outputs; it is not corrected with sampling or repetition penalties.

Interpretation and limits

ROUGE measures reference overlap rather than factual consistency. Low cross-entropy alone cannot establish good summarization because teacher forcing supplies correct preceding words during validation. Greedy generation must recover from its own errors. This is one seed on a small, ordered subset, with a 1,024-example validation subset; no confidence intervals or claim of competitive summarization quality are made. More representative training data, a wider/deeper model, longer training, and additional validation would be the first improvements. Test scores must not guide those choices.

5. Attention recorded during generation

The heatmap uses cross-attention from the last decoder layer for test example 1. Four heads are averaged. Each row was recorded while generating its labelled predicted token; no gold target was fed to the model. The source window contains 50 consecutive subwords starting at index 337, chosen to maximize summed attention over all generated steps. For legibility the figure shows the first 24 generated subwords. Underscores mark SentencePiece word boundaries.



The selected window contains 13.6% of attention mass per generated row on average (over all generated rows). Probabilities are not renormalized after cropping. The complete attention matrix is supplied in results/attention.npz. A large weight indicates model focus, not proof that a source fact caused a correct prediction. The largest individual weight is 0.020997, compared with uniform 1/400 = 0.002500. A uniform 50-token window would hold 12.5% of the mass. Concentration alone does not establish a correct semantic alignment. Repeated rows alongside repeated output can indicate a decoding loop; diffuse attention can indicate weak alignment. Neither pattern alone proves an error mechanism.

6. Qualitative analysis protocol

The next five pages show the first five test examples in fixed repository order. This selection was fixed before evaluating outputs and avoids cherry-picking. Article excerpts are explicitly labelled; references and generated summaries are unedited and complete. A complete per-test prediction file is provided separately without articles. The diagnostics below each example describe repetition and overlap; they do not certify facts.

7.1. Test example 1

Dataset ID: f001ec5c4704938247d27a44948eebb37ae98d01

Article excerpt (first 1,100 characters)

(CNN)The Palestinian Authority officially became the 123rd member of the International Criminal Court on Wednesday, a step that gives the court jurisdiction over alleged crimes in Palestinian territories. The formal accession was marked with a ceremony at The Hague, in the Netherlands, where the court is based. The Palestinians signed the ICC's founding Rome Statute in January, when they also accepted its jurisdiction over alleged crimes committed "in the occupied Palestinian territory, including East Jerusalem, since June 13, 2014." Later that month, the ICC opened a preliminary examination into the situation in Palestinian territories, paving the way for possible war crimes investigations against Israelis. As members of the court, Palestinians may be subject to counter-charges as well. Israel and the United States, neither of which is an ICC member, opposed the Palestinians' efforts to join the body. But Palestinian Foreign Minister Riad al-Malki, speaking at Wednesday's ceremony, said it was a move toward greater justice. "As Palestine formally becomes a State Party to the Rome St

Reference highlights (original newlines)

Membership gives the ICC jurisdiction over alleged crimes committed in Palestinian territories since last June .
Israel and the United States opposed the move, which could open the door to war crimes investigations against Israelis .

Generated summary (unedited greedy output)

new: "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's" "i's" "i's" "i's "i's "i's" "i's "i's "i's" "i's" "i's"

Observed error indicators

The key reference information is ICC jurisdiction and Palestinian membership. Of the diagnostic source-name/topic strings icc, palestin, the output contains none. String presence alone does not establish a correct factual statement.

The unedited output contains 26 whitespace-delimited words; 16 repeated trigram occurrences beyond their first appearances. It shares 0 distinct whitespace tokens with the reference. These counts diagnose repetition and lexical overlap, not factual accuracy. Compare the source excerpt and reference above to check whether the output preserves the event, participants, and quantities. Generic or repeated text does not identify the key facts even when it contains common reference words.

7.2. Test example 2

Dataset ID: 230c522854991d053fe98a718b1defa077a8efef

Article excerpt (first 1,100 characters)

(CNN)Never mind cats having nine lives. A stray pooch in Washington State has used up at least three of her own after being hit by a car, apparently whacked on the head with a hammer in a misguided mercy killing and then buried in a field -- only to survive. That's according to Washington State University, where the dog -- a friendly white-and-black bully breed mix now named Theia -- has been receiving care at the Veterinary Teaching Hospital. Four days after her apparent death, the dog managed to stagger to a nearby farm, dirt-covered and emaciated, where she was found by a worker who took her to a vet for help. She was taken in by Moses Lake, Washington, resident Sara Mellado. "Considering everything that she's been through, she's incredibly gentle and loving," Mellado said, according to WSU News. "She's a true miracle dog and she deserves a good life." Theia is only one year old but the dog's brush with death did not leave her unscathed. She suffered a dislocated jaw, leg injuries and a caved-in sinus cavity -- and still requires surgery to help her breathe. The veterinary hospita

Reference highlights (original newlines)

Theia, a bully breed mix, was apparently hit by a car, whacked with a hammer and buried in a field .
"She's a true miracle dog and she deserves a good life," says Sara Mellado, who is looking for a home for Theia .

Generated summary (unedited greedy output)

new: "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i's "i" "i's "i" "i's "i's "we" "i's "

Observed error indicators

The key reference information is Theia surviving severe injuries, her rescue and care. Of the diagnostic source-name/topic strings theia, dog, the output contains none. String presence alone does not establish a correct factual statement.

The unedited output contains 34 whitespace-delimited words; 23 repeated trigram occurrences beyond their first appearances. It shares 0 distinct whitespace tokens with the reference. These counts diagnose repetition and lexical overlap, not factual accuracy. Compare the source excerpt and reference above to check whether the output preserves the event, participants, and quantities. Generic or repeated text does not identify the key facts even when it contains common reference words.

7.3. Test example 3

Dataset ID: 4495ba8f3a340d97a9df1476f8a35502bcce1f69

Article excerpt (first 1,100 characters)

(CNN)If you've been following the news lately, there are certain things you doubtless know about Mohammad Javad Zarif. He is, of course, the Iranian foreign minister. He has been U.S. Secretary of State John Kerry's opposite number in securing a breakthrough in nuclear discussions that could lead to an end to sanctions against Iran -- if the details can be worked out in the coming weeks. And he received a hero's welcome as he arrived in Iran on a sunny Friday morning. "Long live Zarif," crowds chanted as his car rolled slowly down the packed street. You may well have read that he is "polished" and, unusually for one burdened with such weighty issues, "jovial." An Internet search for "Mohammad Javad Zarif" and "jovial" yields thousands of results. He certainly has gone a long way to bring Iran in from the cold and allow it to rejoin the international community. But there are some facts about Zarif that are less well-known. Here are six: . In September 2013, Zarif tweeted "Happy Rosh Hashanah," referring to the Jewish New Year. That prompted Christine Pelosi, the daughter of House Mino

Reference highlights (original newlines)

Mohammad Javad Zarif has spent more time with John Kerry than any other foreign minister .
He once participated in a takeover of the Iranian Consulate in San Francisco .
The Iranian foreign minister tweets in English .

Generated summary (unedited greedy output)

new: "i's "i's "i's "i's "i's "i" "i" "i's "i" "i" "i's "i's "i's "i" "i" "i" "i" obama"

Observed error indicators

The key reference information is Zarif, diplomacy with John Kerry, and English tweets. Of the diagnostic source-name/topic strings zarif, kerry, the output contains none. String presence alone does not establish a correct factual statement.

The unedited output contains 51 whitespace-delimited words; 38 repeated trigram occurrences beyond their first appearances. It shares 0 distinct whitespace tokens with the reference. These counts diagnose repetition and lexical overlap, not factual accuracy. Compare the source excerpt and reference above to check whether the output preserves the event, participants, and quantities. Generic or repeated text does not identify the key facts even when it contains common reference words.

7.4. Test example 4

Dataset ID: a38e72fed88684ec8d60dd5856282e999dc8c0ca

Article excerpt (first 1,100 characters)

(CNN)Five Americans who were monitored for three weeks at an Omaha, Nebraska, hospital after being exposed to Ebola in West Africa have been released, a Nebraska Medicine spokesman said in an email Wednesday. One of the five had a heart-related issue on Saturday and has been discharged but hasn't left the area, Taylor Wilson wrote. The others have already gone home. They were exposed to Ebola in Sierra Leone in March, but none developed the deadly virus. They are clinicians for Partners in Health, a Boston-based aid group. They all had contact with a colleague who was diagnosed with the disease and is being treated at the National Institutes of Health in Bethesda, Maryland. As of Monday, that health care worker is in fair condition. The Centers for Disease Control and Prevention in Atlanta has said the last of 17 patients who were being monitored are expected to be released by Thursday. More than 10,000 people have died in a West African epidemic of Ebola that dates to December 2013, according to the World Health Organization. Almost all the deaths have been in Guinea, Liberia and Si

Reference highlights (original newlines)

17 Americans were exposed to the Ebola virus while in Sierra Leone in March .
Another person was diagnosed with the disease and taken to hospital in Maryland .
National Institutes of Health says the patient is in fair condition after weeks of treatment .

Generated summary (unedited greedy output)

new: new: "i-old" new: "i-old, police say .
new: "i-old, police say .
new: "i-old, police say .
new: "i-old, police say .
new: "weed in the u.s.s.s

7.5. Test example 5

Dataset ID: c27cf1b136cc270023de959e7ab24638021bc43f

Article excerpt (first 1,100 characters)

(CNN)A Duke student has admitted to hanging a noose made of rope from a tree near a student union, university officials said Thursday. The prestigious private school didn't identify the student, citing federal privacy laws. In a news release, it said the student was no longer on campus and will face student conduct review. The student was identified during an investigation by campus police and the office of student affairs and admitted to placing the noose on the tree early Wednesday, the university said. Officials are still trying to determine if other people were involved. Criminal investigations into the incident are ongoing as well. Students and faculty members marched Wednesday afternoon chanting "We are not afraid. We stand together," after pictures of the noose were passed around on social media. At a forum held on the steps of Duke Chapel, close to where the noose was discovered at 2 a.m., hundreds of people gathered. "You came here for the reason that you want to say with me, 'This is no Duke we will accept. This is no Duke we want. This is not the Duke we're here to experi

Reference highlights (original newlines)

Student is no longer on Duke University campus and will face disciplinary review .
School officials identified student during investigation and the person admitted to hanging the noose, Duke says .
The noose, made of rope, was discovered on campus about 2 a.m.

Generated summary (unedited greedy output)

new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .
new: "we, police say .

Observed error indicators

The key reference information is Duke student disciplinary review following the noose incident. Of the diagnostic source-name/topic strings duke, noose, the output contains none. String presence alone does not establish a correct factual statement.

The unedited output contains 75 whitespace-delimited words; 68 repeated trigram occurrences beyond their first appearances. It shares 1 distinct whitespace tokens with the reference. These counts diagnose repetition and lexical overlap, not factual accuracy. Compare the source excerpt and reference above to check whether the output preserves the event, participants, and quantities. Generic or repeated text does not identify the key facts even when it contains common reference words.

8. Collapse investigation and correction

The original checkpoint was audited on validation indices 0, 1, 2, 100, 500 and 1000. All six source token sequences were different. Original encoder layer 1 retained differences (RMS 0.86-0.87), but layer 2 reduced them to about 0.0013. Fresh random weights produced encoder differences around 1.36-1.37. This is evidence of learned representation collapse rather than repeated tokenized input.

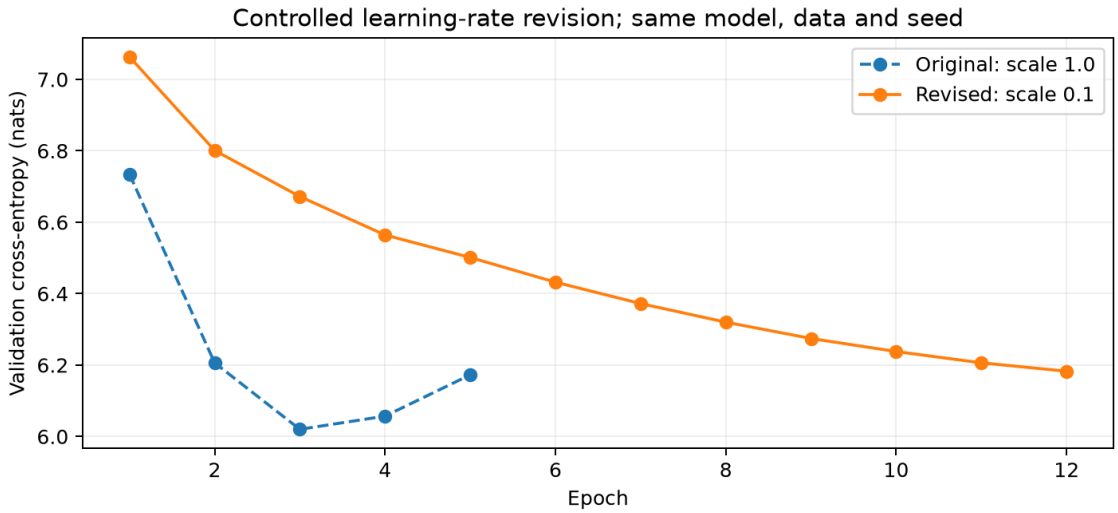
Cached and uncached greedy IDs matched exactly for all six real validation articles, including per-article versus batch runs and a shortened validation source to exercise padding. Hooks checked both decoder layers at every generation step: encoder K/V values and source masks were correct; blocked positions had zero attention. Cached versus full-prefix logit error was at most 5.7e-06. The complete original validation loss recomputed to exactly 6.0200552, matching the checkpoint. No decoder routing or cache defect was found.

The original schedule peaked at 0.0088388, which is aggressive for this small post-normalization model. A controlled retraining uses a 0.1 scale (peak 0.00088388), a maximum of 12 epochs and the same patience of 2. No architecture or decoding changes are introduced to force diverse outputs. The full audit is repeated on the revised validation checkpoint.

Validation diagnostic	Original	Revised
Distinct input sequences / 6	6	6
Distinct generated sequences / 6	1	6
Encoder RMS: article 1 vs 2	0.001306	1.445921
Cached = uncached	True	True
Recomputed validation loss	6.020055	6.182745

The original data/tokenizer reuse was valid, but the scripts failed to check it. The corrected scripts validate counts, unique/disjoint IDs, portable content hashes, BPE vocabulary, special token IDs and tokenizer fingerprints. Fresh streams from pinned revision 96df5e686bee6baa90b8bee7c28b81fa3fa6223d matched all three saved subsets. Training-only tokenizer reconstruction reproduced every piece and score.

9. Stability comparison



The comparison above uses validation only. The original run can have a lower teacher-forced loss while ignoring its source: the correct preceding target words still help next-token prediction. Restoring source sensitivity does not itself guarantee a lower loss, fluent generation or accurate facts. The revised checkpoint is chosen by the lowest validation loss within the revised, fixed training run. The unchanged small data sample remains a major limitation.

10. Verification and reproduction

Unit tests verify future-token isolation, source and target padding masking, attention normalization, ignored padding loss, summary shifting, sinusoidal positions, optimization on a fixed mini-batch, sentence boundaries, and equality of cached and full-prefix greedy predictions. See test_results.txt for the recorded run and README.md for commands.

Observed host: Python 3.14.8 on Windows 11; PyTorch runtime 2.14.1+cpu (distribution metadata 2.14.1), NumPy 2.5.4 in the local dependency overlay, datasets 5.1.0, SentencePiece 0.2.2, rouge-score 0.1.2, matplotlib 3.11.2, reportlab 5.1.0, nltk 3.10.4 and protobuf 7.36.2. The original training run used base NumPy 2.5.3; this revised run uses 2.5.4. PyTorch is unchanged. A separate virtual environment with these pinned direct dependencies and system-site-packages passed pip check, all 14 tests, dataset verification and the offline downloader reuse path. A fully isolated clean-machine installation is not claimed.

The archive includes source scripts, this report, measured metrics, training history, figures, five qualitative examples, test predictions and generation-time attention. It excludes dataset downloads, model checkpoints, the tokenizer training corpus and installed dependencies. Tokenizer and model artifacts are regenerated in a separate work directory.

References

Assignment: NLP Group Assignment 2, DSL 504, Question 3, pages 8-11.

Vaswani et al. (2017). Attention Is All You Need. https://arxiv.org/abs/1706.03762

CNN / Daily Mail 3.0.0, canonical Hugging Face repository: https://huggingface.co/datasets/abisee/cnn_dailymail

SentencePiece: https://github.com/google/sentencepiece

ROUGE implementation: https://github.com/google-research/google-research/tree/master/rouge

PyTorch softmax documentation: https://docs.pytorch.org/docs/stable/generated/torch.nn.Softmax.html

PLOT-TWIST GENERATOR

Comparing Vanilla RNN and LSTM Language Models

Course: Natural Language Processing

Assignment: Group Assignment 2 — Question 4

Dataset: TinyStories (selected 22,000-story stream subset)

Models: Vanilla RNN and LSTM | Seed: 42

Submission status: experiments and outputs are included. Genuine human ratings are still required; blank rating forms are provided and are not represented as completed evaluation.

Abstract

This project implements two recurrent neural language models—a vanilla Recurrent Neural Network (RNN) and a Long Short-Term Memory (LSTM)—to continue short stories from prompts. Both models use the same word-level vocabulary and story-level train/validation/test split. They are trained with next-token cross-entropy and compared using held-out test perplexity, lexical diversity, 3-gram repetition, and qualitative inspection of a Plot-Twist Challenge. The recorded LSTM checkpoint achieves lower test perplexity (13.40 versus 16.88), higher Distinct-1/Distinct-2, and a lower mean 3-gram repetition rate than the RNN. However, both models often drift away from the unusual prompt and produce generic children's-story events rather than meaningful twists. The results demonstrate that stronger next-token prediction and higher lexical diversity do not guarantee coherent narrative planning. A blinded human-rating set is prepared, but its ratings remain blank pending genuine evaluation.

1. Objectives

- Prepare a consistent tokenization and next-token prediction pipeline.
- Train a vanilla RNN and an LSTM under comparable settings.
- Compare validation/test perplexity, diversity, repetition, and output length.
- Implement greedy decoding, temperature sampling, and top-k sampling.
- Test at least five twist-oriented openings and inspect failures rather than omitting them.
- Prepare a blinded human-evaluation form; do not treat uncollected ratings as results.

2. Dataset and preprocessing

The code uses the TinyStories dataset (`roneneldan/TinyStories`) through the Hugging Face `datasets` library. It takes the first 22,000 stories from the training stream, shuffles them with seed 42, and divides them at the story level into 20,000 training stories, 1,000 validation stories, and 1,000 test stories. The raw dataset and processed token-ID splits are intentionally excluded from the ZIP; rerunning preprocessing requires internet access to download the dataset.

Setting	Implementation
Tokenization	Lowercased word-level regex; curly quotation marks normalized
Vocabulary	Built from training split only; minimum frequency 5; maximum 10,000 entries
Saved vocabulary	6,695 tokens, including four special tokens
Special tokens	<pad>=0, <unk>=1, <bos>=2, <eos>=3
Sequence length	Maximum 200 tokens; longer stories truncated
Story boundaries	Each story encoded independently; no concatenation across stories
Padding and loss	Batch padding; cross-entropy ignores <pad> targets

Split	Stories	Tokens	Mean length	UNK rate	Truncated
Train	20,000	3,483,783	174.2	0.22%	32.7%
Validation	1,000	173,213	173.2	0.34%	30.7%
Test	1,000	174,787	174.8	0.34%	32.2%

A limitation of this preprocessing run is that roughly 31–33% of stories are truncated at 200 tokens; when truncation occurs, an end-of-sequence token is not appended. The recorded vocabulary and generated samples also reveal a small number of garbled symbols (for example, "€" and "™"), indicating that text cleaning could be improved.

3. Model architecture and training

Both models use an embedding layer, one recurrent layer, and a linear projection to vocabulary logits. The RNN uses `torch.nn.RNN`; the LSTM uses `torch.nn.LSTM`. Both are trained to predict the next token from the preceding token sequence using cross-entropy loss, with padding excluded from the loss. Model comparison is controlled by using the same vocabulary, split, embedding/hidden dimensions, batch size, optimizer, and seed.

Setting	Value
Embedding / hidden dimension	256 / 512

Recurrent layers	1
Dropout	0.2
Optimizer / learning rate	Adam / 0.001
Batch size	64
Gradient clipping	1.0
Maximum epochs / patience	10 / 3
Random seed	42
Device	Original run hardware not recorded in the available artifacts

Both recorded training histories contain 10 epochs, and validation loss was still decreasing at the final epoch. Early stopping therefore did not trigger, and both checkpoints should be viewed as potentially undertrained. The loss curves are provided in `artifacts/loss_curves.png`; detailed epoch histories are saved in `artifacts/rnn_history.json` and `artifacts/lstm_history.json`.

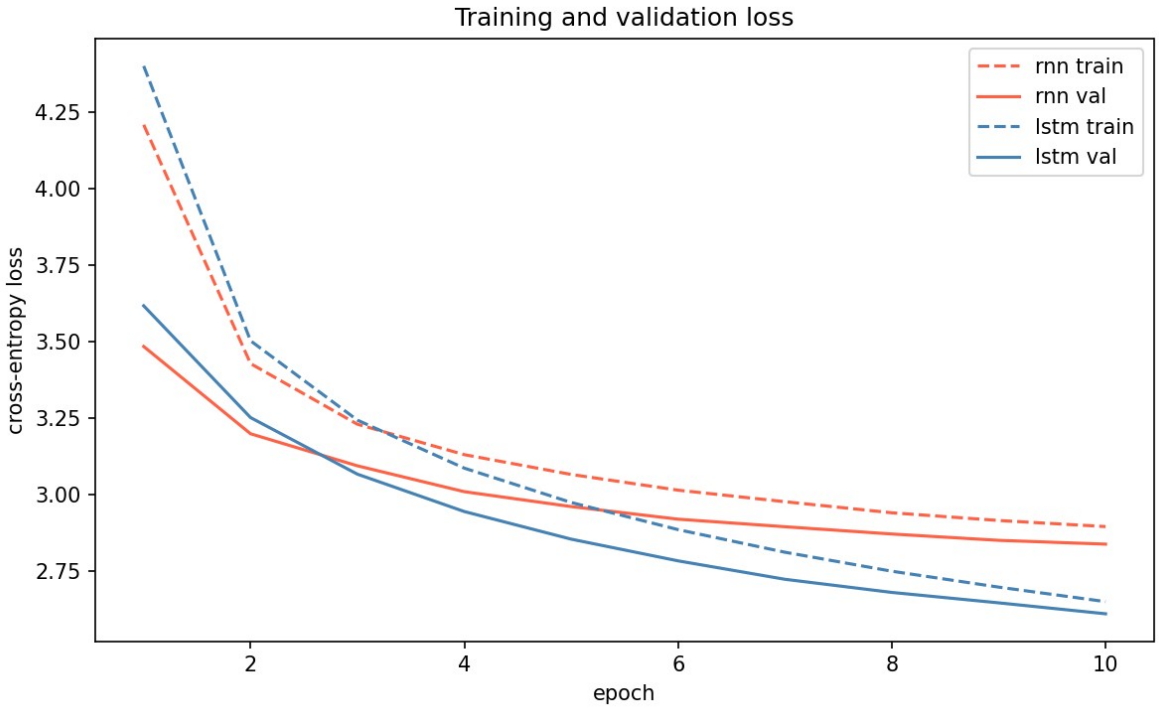


Figure 1. Recorded training and validation loss curves for the two models.

4. Quantitative results

Metric	Vanilla RNN	LSTM
Trainable parameters	5,542,695	6,725,415
Training time	343.74 s	473.14 s
Best validation perplexity	17.07	13.59
Test loss	2.8261	2.5955
Test perplexity	16.88	13.40
Distinct-1	0.189	0.205
Distinct-2	0.480	0.544
Mean 3-gram repetition rate	0.1033	0.0837
Mean output length, including prompt	70.77 words	78.91 words
Generation samples for pooled metrics	35	35

Perplexity is exp(mean per-token cross-entropy); lower is better for next-token prediction on the same held-out data. Distinct-1/2 are unique unigrams/bigrams divided by total unigrams/bigrams. Repetition rate is 1 − (unique 3-grams / total 3-grams), averaged per output. The pooled metrics use 35 generated outputs per model from seven general prompts and five decoding settings; the prompt is included in these pooled texts. Under the project's thresholds, no outputs were under 20 words and none reached 180 words. The latter count is not very informative because generation is capped at 80 new tokens.

The LSTM has about 21.3% more parameters than the RNN and took about 37.6% longer in the recorded run. Despite that cost, it improved test perplexity from 16.88 to 13.40, raised Distinct-2 from 0.480 to 0.544, and reduced mean 3-gram repetition from 0.1033 to 0.0837. These are consistent improvements on the measured tasks, but they do not directly measure plot quality.

5. Generation strategies and Plot-Twist Challenge

Strategy	Rule	Observed trade-off
Greedy	Choose the highest-probability next token each step.	Deterministic and often predictable; can fall into repetitive loops.
Temperature sampling	Divide logits by T, then sample; tested T=0.5, 0.8, 1.2.	Higher T increases variation but can produce abrupt or incoherent transitions.
Top-k sampling	Sample among the k most likely tokens; k=10.	Restricts unlikely tokens but does not enforce character or plot consistency.

The Plot-Twist Challenge contains 84 saved records: seven prompts × two models × three strategies × two sample slots. It covers an exam paper with tomorrow's answers, a robot receiving a message from its future self, a detective solving a future murder, a village forgetting the previous day, a letter from 100 years in the future, a new student who knows everyone's secrets, and the moon moving closer to a village. Greedy decoding is deterministic for the same prompt and checkpoint, so its paired records are duplicates. Outputs are preserved without rewriting in `results/plot_twist_challenge.json`.

6. Qualitative analysis

6.1 RNN greedy: prompt drift and repetition

Prompt: "A detective solved a murder that had not happened yet."

Recorded continuation (verbatim excerpt): "he was very excited and wanted to play with it . he asked his mom , " what is that ? " the bird said , " i am a little girl . i am a little girl . i am a little girl . i am a little girl . i am a little girl . i am a little girl . i am a little girl . i am a little girl"

The output quickly abandons the detective and the future-murder premise. The shift to a bird and a little girl is unmotivated, and the repeated sentence is a clear degeneration loop. No investigation or twist develops.

6.2 RNN temperature: more variety without relevance

Prompt: "A student found an exam paper containing tomorrow's answers."

Recorded continuation: "he was a very powerful puppy who was very competitive and he loved to explore the world . one day , he saw a big lake with many things in a cage . the cat wanted to eat the food but it was too late . timmy and his friends were excited to have found their puppy ."

The sample includes more entities and events than the greedy loop, but the exam paper and future answers disappear. The puppy, lake, cat, and Timmy do not form a coherent chain tied to the prompt. More varied text is not necessarily a better continuation.

6.3 LSTM greedy: locally readable but generic

Prompt: "A student found an exam paper containing tomorrow's answers."

Recorded continuation: "the little girl was so excited ! she ran around the house , looking for something to eat . she saw a big , green frog . she was so excited ! she ran to the kitchen and opened it . she was so excited ! she ran around the house and found a big , shiny object ."

The actions have some local continuity, but the continuation ignores the exam premise and repeats "she was so excited". This is a useful example of how locally readable sentences can still fail the story task.

6.4 LSTM temperature: entity drift and malformed text

Prompt: "An old robot received a message from its future self."

Recorded continuation: “she was so excited that she thanked her mom for making her salad for her special day . she said goodbye to the puppy and to the big stack . the little girl was so happy . she shouted the dress - <unk> , and all the books and showed her how to make her own secrets .”

The robot and future message disappear; the output jumps among a girl, salad, puppy, books, and a dress, and contains an unresolved <unk> token. Sampling creates lexical variety but not a consistent narrative.

6.5 Main failure patterns

- Prompt drift: the model falls back to common TinyStories-style scenes rather than using the distinctive premise.
- Character and entity drift: robots, detectives, students, animals, and children change without explanation.
- Repetition: greedy decoding can repeat clauses or sentences, especially for the RNN.
- Weak causal structure: actions and objects appear in sequence without a meaningful relationship.
- Data/tokenization artifacts: <unk> and occasional garbled symbols appear in generated text.
- No reliable twist: the model rarely turns the opening premise into a surprising but consistent development.

Possible future remedies include more training data, longer training with monitored validation loss, improved cleaning and tokenization, repetition penalties or n-gram blocking, and explicit story-planning or stronger prompt-conditioning methods. These interventions were not tested in the recorded experiments and are recommendations for future work, not measured improvements.

7. Human evaluation

The assignment requests at least ten generated continuations rated blindly on coherence, grammar, prompt relevance, and creativity/surprise using a 1–5 scale; three evaluators are recommended where feasible. The package now includes a 30-sample blinded sheet, a 90-row three-rater template, a separate sample-to-model key, and instructions. The score fields are blank because genuine evaluator responses were not available when this report was prepared. No human means, inter-rater agreement, or human preference is claimed.

Before final submission, have real evaluators independently rate the blinded samples. Keep `results/human_eval_key.csv` away from evaluators until scoring is finished. Then calculate per-criterion averages, report the number of raters, discuss disagreements, and compare those ratings with the automatic metrics. This is a rubric-critical outstanding step; the blank form is preparation for the study, not evidence that it has been completed.

8. Discussion: answering the assignment questions

RNN versus LSTM. The LSTM performed better on held-out next-token prediction and had lower pooled repetition, at the cost of more parameters and training time. Its gating mechanism is designed to retain information over longer sequences, but the outputs show that this did not solve narrative planning.

Which generates more coherent, less repetitive text? The recorded LSTM metrics favor it, and representative samples can be locally more readable. However, both models often ignore the prompt; without genuine human ratings, a broad human-perceived coherence claim cannot be made.

Temperature effects. Lower temperature concentrates probability on likely tokens; higher temperature increases variation. The recorded high-temperature examples can drift or become malformed. Temperature 0.8 and top-k are reasonable settings to inspect, but neither guarantees relevance.

Do diversity scores guarantee better stories? No. Diversity counts lexical variety, not meaning, causality, prompt adherence, or surprise. A continuation can be varied and still incoherent.

Does lower perplexity imply better storytelling? No. LSTM perplexity is lower, but both models fail to develop many challenge premises. Perplexity evaluates next-token prediction on held-out text rather than narrative quality directly.

Most common failures and potential fixes. Prompt drift, repetitive loops, character inconsistency, abrupt transitions, and tokenization artifacts. Potential fixes are listed above but were not tested.

Limitations. A 20k/1k/1k story split, one random seed, a 10-epoch cap, truncation of many stories, word-level tokenization, seven challenge prompts, capped generations, and no completed human ratings. Results are therefore exploratory and should not be generalized beyond this setup.

9. Reproducibility and submission contents

The archive includes source scripts, model checkpoints, the vocabulary, data statistics, loss histories and plot, saved quantitative metrics, general generation outputs, challenge outputs, and human-rating materials. It excludes the raw dataset and processed split tensors to avoid bundling the dataset subset. Preprocessing and test-set reevaluation require downloading TinyStories again.

- Install dependencies: `pip install -r requirements.txt`.
- Download/preprocess data: `python -m src.preprocess`.
- Train either model: `python -m src.train --model rnn` or `python -m src.train --model lstm`.
- Regenerate loss plot: `python -m src.plot_curves`.
- Generate continuations: `python -m src.generate --model lstm --strategy temperature --temperature 0.8`.
- Run the challenge: `python -m src.challenge`.
- Recompute metrics: `python -m src.metrics` (requires processed train/validation/test splits).
- Create blinded evaluation forms: `python -m src.human_eval`.

The run hardware and exact installed library versions were not preserved in the supplied artifacts; training times are therefore reported as recorded, not as a portable runtime guarantee.

10. Conclusion

In this run, the LSTM is the stronger next-token language model: it achieves lower test perplexity, somewhat higher lexical diversity, and lower mean 3-gram repetition than the vanilla RNN. Yet both models struggle with the core creative task. Their continuations frequently replace the challenge premise with generic children's-story events, and greedy decoding can loop. The central conclusion is that next-token modeling metrics and lexical diversity are useful but insufficient measures of storytelling ability. A genuine blinded human evaluation remains necessary before the final submission can claim to have completed the assignment's human-evaluation requirement.

Experimental summary and limitations

Question	Experiment and result	Limitations
Q1: translation	50,000 training pairs; LSTM baseline versus additive attention; test BLEU 0.60 versus 0.92	One seed per model; word-level vocabulary; filtered test subset; low translation quality
Q2: review ratings	200,000 training reviews; two seeds; test Macro F1 0.6351 for BiGRU versus 0.6706 for RoBERTa	Two seeds; sequence truncation; mixed training topology; pretrained versus non-Transformer comparison
Q3: summarization	2,048 training examples; 11,490 test articles; revised model ROUGE-1 1.7506 versus Lead-3 39.9077	Small ordered subset; pervasive repetition; reused test benchmark; no competitive quality claim
Q4: story generation	20,000 training stories; RNN/LSTM test perplexity 16.88/13.40; seven challenge prompts	Prompt drift and repetition; pooled metrics include prompts; human-rating results not yet available

Q4 length measurement: original pooled metrics include prompt text and use an excessive-length threshold beyond the 80-new-token cap. Supplemental analysis of the separately saved general-generation corpus gives continuation-only means of 63.43 words (RNN) and 67.69 (LSTM), with 24/35 and 33/35 reaching 80 generated tokens. These are separate-corpus, descriptive cap counts; they are not replacements for original pooled metrics or evidence of semantic incompleteness.

Across the experiments, attention and pretraining improve some measured tasks but do not guarantee fluent generation, factual accuracy or narrative coherence. Next-token loss and overlap/diversity metrics address different aspects of performance. Qualitative examples and explicit error analysis are needed to interpret the numerical results.

The supplied source, configurations, executed notebooks, predictions, figures and model-specific report sections document the experiment settings. Reproduction instructions are provided in the main README and each question directory. Dataset/model revisions and package versions are recorded where available; numerical equivalence across environments is not guaranteed.